flexyBayes – exact aggregation for out-of-core mixed models

A numerical study, with technical appendices on the mathematics, prior art, and limits

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Contents

1 What was built **1**

2 The numerical study – per-row versus aggregated **2**

3 Scaling and the maximum realistic N **3**

4 When the approach breaks – the complexity envelope **3**

5 Appendix A – Mathematics **4**
5.1 A.1 Cells and sufficient statistics 4
5.2 A.2 The per-family aggregated log-likelihood 4
5.3 A.3 Floating-point precision and the exactness ceiling 4

6 Appendix B – Prior art and references **5**

7 References **5**

8 Appendix C – The complexity envelope **6**

Summary. flexyBayes fits its core mixed-model family – gaussian, binomial, and poisson generalized linear mixed models with factor fixed effects and random intercepts – to datasets that do not fit in memory, by reading the data in chunks and accumulating exact per-cell sufficient statistics. The fitted posterior is the full-data posterior, not an approximation. The numerical study below fits the same models per row and by aggregation and finds the fixed-effect estimates agree to about 10^{-10} and the variance components to between 10^{-7} (count families) and 10^{-4} (the gaussian residual scale, where a custom prior is composed). Memory is bounded by one chunk plus the cell accumulator, not by the row count: a five-billion-row fit used about 846 MB. **The method is not new** – it is the exponential-family sufficient-statistic reduction combined with covariate-pattern aggregation, with direct streaming precedent in Lumley's `biglm` – applied here as an automatic, refusal-gated, multi-backend Bayesian fitting path (Appendix B). **Maximum realistic N:** the algorithm is memory-unbounded; exactness holds to roughly 10^{15} rows for well-conditioned data; the practical ceiling on a 32 GB / 1.3 TB workstation is about 10^{11} rows (disk- and time-bound), and the true binding constraint is the cell count K, not N (Section 3, Appendix A).

1 What was built

Two entry points share one estimation path. `flexybayes_stream()` reads an `.fst` file, a set of partitioned `.fst` shards, a delimited file, an in-memory frame, or a chunk generator, accumulates per-cell sufficient

statistics across chunks, and fits the cell-level model through INLA or greta. `flexybayes(aggregate = "auto")` reaches the same aggregation when the data fits in memory. Supported families are gaussian (identity), binomial (logit), and poisson (log); supported structure is factor fixed effects and random-intercept grouping. Everything else is refused before any data is read (Section 4, Appendix C).

2 The numerical study – per-row versus aggregated

The same model, `y ~ env + (1 | geno)` on a four-environment by thirty-genotype design (120 cells), was fitted two ways on identical data with matched priors: the ordinary per-row path (`aggregate = FALSE`) and the exact-aggregated path (`aggregate = TRUE`), both through INLA's deterministic Laplace engine, so any difference is numerical rather than Monte-Carlo. Sample sizes run from moderate (10^4) to the border of per-row feasibility (10^6; beyond about 2×10^6 the per-row INLA fit exhausts a 32 GB machine, Section 3).

The per-cell estimates at $N = 10^6$ are reproduced side by side; the two paths return the same numbers.

Table 1: Gaussian fit, N = 1,000,000 (120 cells): every parameter, both paths.

parameter	per row	aggregated	|difference|
(Intercept)	-0.127209	-0.127209	0.00e+00
env2	1.194523	1.194523	0.00e+00
env3	-0.700538	-0.700538	0.00e+00
env4	0.502068	0.502068	0.00e+00
sigma (residual SD)	1.499779	1.499778	1.60e-06
tau (geno SD)	0.692873	0.692893	2.02e-05

Across all three families and all sample sizes, the largest disagreement in any fixed-effect coefficient and in any variance component is summarised below.

Table 2: Largest per-path difference, by family and sample size.

family	N	K	max|beta diff|	max|var-comp diff|
binomial	10,000	120	1.3e-07	1.9e-07
binomial	100,000	120	1.2e-10	4.6e-08
binomial	1,000,000	120	2.0e-10	8.8e-08
gaussian	10,000	120	7.4e-08	1.7e-04
gaussian	100,000	120	7.8e-08	6.6e-04
gaussian	1,000,000	120	1.5e-09	2.0e-05
poisson	10,000	120	7.7e-09	2.4e-07
poisson	100,000	120	1.3e-09	1.2e-07
poisson	1,000,000	120	1.5e-11	1.6e-07

What the comparison shows. The fixed-effect coefficients agree to about 10^{-10} in every family and at every size: the aggregated likelihood is the same likelihood (Appendix A.2), so the linear-predictor estimates are identical to numerical precision. The variance components for the count families agree to about 10^{-7} – INLA is given the cell totals directly, with no auxiliary construction. The one place a larger difference appears is the gaussian residual and group standard deviations, which differ by up to about 10^{-4}. That residual is explained, not by the aggregation, but by how the within-cell sum of squares is supplied to INLA on the gaussian path: it enters through a custom log-precision prior composed with INLA's own precision prior, and the composition is not bit-identical to the per-row prior treatment (Appendix A.4). The difference is small, it does not touch the fixed effects, and the count families – which need no such construction – do not show it.

3 Scaling and the maximum realistic N

Because only one chunk and the K-cell accumulator are ever resident, peak memory does not grow with N. Partitioned `.fst` shards were streamed on a single 32 GB machine on a 1200-cell design:

Table 3: Streaming a partitioned .fst dataset; gaussian, INLA. Peak memory is the OS resident size of a fresh process.

N	K	wall (s)	peak (MB)	on disk (GB)	intercept
10,000,000	1200	3.3	708	0.09	0.98542
100,000,000	1200	5.7	796	0.94	0.98552
1,000,000,000	1200	31.9	844	9.40	0.98565
5,000,000,000	1200	150.9	846	46.98	0.98568

Peak memory moved from 708 MB at ten million rows to 846 MB at five billion – a five-hundred-fold increase in data through a roughly flat envelope – and the recovered intercept was stable throughout. For contrast, the per-row path reached about 12 GB at one million rows and was infeasible at five million on the same machine.

The maximum realistic N. Three different ceilings apply, and they are far apart:

- *Algorithm (memory).* Unbounded. Peak memory is set by `chunk_rows` and K, not N.
- *Exactness (floating point).* Counts and sums are IEEE-754 doubles, exact for integers to $2^{53} \approx 9.0\times10^{15}$. For order-one response values the sums stay exact to roughly $N \approx 10^{15}$ rows; the within-cell sum-of-squares loses precision only when the within-cell signal-to-noise ratio is extreme, independent of N (Appendix A.3).
- *Practical (disk and time).* On this 32 GB / 1.3 TB workstation a three-column dataset is about 9.4 GB per billion rows, so disk caps a single machine near 1.3×10^{11} rows, read in roughly an hour. A cluster with more disk reaches the exactness ceiling.

The binding constraint in practice is the cell count K, not the row count N. A replicated factorial design fixes K by its factor combinations, so N may grow to the precision ceiling; a design whose cell count grows with N does not compress at all and is refused (Section 4).

Recommendation for the package. Do not impose a hard cap on N: the method scales, and an arbitrary cap would be misleading. Keep `chunk_rows` at its 5×10^6 default (a few hundred megabytes per chunk, a good throughput-versus-memory balance). Document the path as validated to 5×10^9 rows (measured here) and exact to roughly 10^{13} rows for well-conditioned data, and emit an informational note above about 10^{12} rows recommending the numerically stable centred-sum accumulation (or a rescaled response) when the within-cell signal-to-noise ratio is large. The number worth surfacing to the user is the compression ratio K/N, which the preflight already reports, not a row-count limit.

4 When the approach breaks – the complexity envelope

Exact aggregation applies only while every row in a cell shares one linear predictor and the likelihood reduces to a finite per-cell statistic. As model complexity rises the method degrades predictably; each step beyond the envelope is refused with a named reason code rather than silently approximated. The ladder is tabulated in Appendix C; in brief: factor fixed effects and random intercepts aggregate exactly; a continuous fixed covariate gives one cell per distinct value and so does not compress; random slopes, structured covariance, and smooth terms make the linear predictor vary within a cell and break the reduction; non-canonical links and non-exponential-family likelihoods have no additive sufficient statistic. For any of these the per-row `flexybayes()` path fits the model unchanged.

5 Appendix A – Mathematics

5.1 A.1 Cells and sufficient statistics

For observation i let the linear predictor be $\eta_i = x_i^\top \beta + \sum_j z_{ij} u_j$, with β the fixed effects and u_j the random-intercept effects. Define a *cell* k as a maximal set of observations sharing the same fixed-effect design row and the same random-effect grouping levels; by construction every row in cell k has the same linear predictor η_k. The likelihood factorises over cells, and within a cell of n_k conditionally independent observations the exponential-family log-likelihood is

$$\ell_k = \sum_{i \in k} \Big[\tfrac{y_i \theta_k - b(\theta_k)}{\phi} + c(y_i, \phi) \Big] = \tfrac{\theta_k \sum_{i \in k} y_i - n_k\, b(\theta_k)}{\phi} + \sum_{i \in k} c(y_i, \phi),$$

where θ_k is the natural parameter (a function of η_k). The parameter-dependent part depends on the data only through n_k and $\sum_{i \in k} y_i$. The term $\sum_{i \in k} c(y_i, \phi)$ is free of the mean parameters; it is dropped for the count families (dispersion $\phi = 1$) and retained for the gaussian, where $\phi = \sigma^2$ is itself a parameter (A.2).

5.2 A.2 The per-family aggregated log-likelihood

Write $S_k^{(1)} = \sum_{i \in k} y_i$ and $S_k^{(2)} = \sum_{i \in k} y_i^2$.

Gaussian, identity link. With $\mu_k = \eta_k$,

$$\ell = -\tfrac{N}{2} \log(2\pi\sigma^2) - \tfrac{1}{2\sigma^2} \sum_k \big[\, S_k^{(2)} - 2\mu_k S_k^{(1)} + n_k \mu_k^2 \,\big],$$

which is algebraically identical to $\sum_i \big[-\tfrac{1}{2} \log(2\pi\sigma^2) - (y_i - \mu_i)^2/(2\sigma^2) \big]$. The cell needs $(n_k, S_k^{(1)}, S_k^{(2)})$; the σ^2-coupled $S_k^{(2)}$ must be retained, so the residual-variance posterior is exact in the likelihood.

Poisson, log link. With exposure E_i and $\lambda_i = E_i \exp(\eta_k)$, the cell contribution is, up to the parameter-free $\sum(y_i \log E_i - \log y_i!)$,

$$S_k^{(1)}\, \eta_k - \exp(\eta_k) \textstyle\sum_{i \in k} E_i.$$

Equivalently, independent Poissons with a shared rate add, so the cell total $S_k^{(1)} \sim \text{Poisson}\big(\exp(\eta_k) \sum_{i \in k} E_i \big)$; INLA is given the response $S_k^{(1)}$ with offset $\sum_{i \in k} E_i$.

Binomial, logit link. With trials m_i and probability p_k, the cell contribution is, up to the parameter-free $\sum \log \binom{m_i}{y_i}$,

$$S_k^{(1)}\, \log p_k + \Big(\textstyle\sum_{i \in k} m_i - S_k^{(1)} \Big) \log(1 - p_k),$$

and independent binomials with shared p_k add, so $S_k^{(1)} \sim \text{Binomial}\big(\sum_{i \in k} m_i,\, p_k \big)$; INLA is given the response $S_k^{(1)}$ with $\sum_{i \in k} m_i$ trials. A 0/1 response is the Bernoulli case $m_i = 1$.

Exactness. By the Fisher–Neyman factorisation theorem the cell statistics are sufficient for the cell parameters, so the aggregated and per-row log-likelihoods differ by an additive, parameter-free constant. The posterior is therefore identical for any prior – this is exact compression, not approximation. Each statistic is a sum, so the dataset total is the sum of the per-chunk totals, which is what makes the accumulation single-pass and bounded-memory.

5.3 A.3 Floating-point precision and the exactness ceiling

Counts and sums are accumulated as IEEE-754 doubles, exact for integers up to $2^{53} \approx 9.0 \times 10^{15}$. While n_k, $S_k^{(1)}$, and $S_k^{(2)}$ stay below that magnitude they are exact; for order-one response values this holds to roughly $N \approx 10^{15}$ rows. (The partitioned-source row total is also carried as a double, not a 32-bit integer, so it does not cap at 2^{31}.)

The within-cell sum of squares $\mathrm{WSS}_k = S_k^{(2)} - (S_k^{(1)})^2/n_k$ is a difference of like-sized quantities and so is subject to cancellation. For a cell with mean μ_k and within-cell standard deviation σ_k, the subtracted terms are of order $n_k\mu_k^2$ while the result is of order $n_k\sigma_k^2$, so the relative error is of order $\varepsilon\,(\mu_k/\sigma_k)^2$, with $\varepsilon \approx 2.2\times10^{-16}$. This depends on the within-cell signal-to-noise ratio, **not** on n_k or N, and is negligible unless μ_k/σ_k is extreme; a numerically stable centred-sum (Welford / Chan) accumulation removes it entirely if a dataset ever requires it.

Finally, the gaussian path injects WSS into INLA through a custom log-precision prior whose constants scale with N (its log-density reaches about $-N/2$ at its mode). At extreme N this destabilises INLA's hyper-parameter integration; recentring the log-density to approximately zero at its mode (an additive constant, absorbed in the prior's normalisation, so small-N fits are unchanged) restores stability. The residual of order 10^{-4} in the gaussian variance components (Section 2) is the imprint of this prior composition, not of the aggregation.

6 Appendix B – Prior art and references

This method is not newly invented. It combines two long-established ideas. First, the exponential-family sufficient-statistic reduction (Fisher 1922; McCullagh and Nelder 1989): the likelihood depends on the data only through a finite statistic, so observations sharing a linear predictor can be summarised without loss – the *covariate-pattern* or *grouped-data* representation that is standard for binomial and poisson regression (McCullagh and Nelder 1989). Second, single-pass bounded-memory fitting of regression models from data larger than memory has direct precedent in Lumley's `biglm` / `bigglm`, which accumulates regression sufficient statistics chunk by chunk (its gaussian path uses the incremental least-squares update of Miller 1992).

What flexyBayes adds is engineering rather than statistics: an automatic, refusal-gated, exact-aggregation path inside a multi-backend *Bayesian* package (greta and INLA), which streams from partitioned files, handles the gaussian within-cell sum of squares through a custom precision prior, and treats exactness and explicit refusal as a first-class contract. The compression is **exact** when a cell shares one linear predictor, which distinguishes it from approximate large-data compression such as *data squashing* (DuMouchel et al. 1999), a lossy moment-matching construction, and from data-partitioning Bayesian schemes such as the consensus Monte Carlo algorithm (Scott et al. 2016), which split the data and recombine sub-posteriors rather than compress losslessly. The deterministic engine used for the exact comparisons here is INLA (Rue, Martino and Chopin 2009).

7 References

- Fisher, R. A. (1922). On the mathematical foundations of theoretical statistics. *Philosophical Transactions of the Royal Society A*, 222, 309–368.
- McCullagh, P. and Nelder, J. A. (1989). *Generalized Linear Models*, 2nd ed. Chapman & Hall/CRC. (Grouped / covariate-pattern data for binomial and poisson regression.)
- Lumley, T. *biglm: Bounded Memory Linear and Generalized Linear Models*. R package; CRAN. (Chunked, bounded-memory regression via accumulated sufficient statistics.)
- Miller, A. J. (1992). Algorithm AS 274: Least squares routines to supplement those of Gentleman. *Applied Statistics*, 41(2), 458–478.
- DuMouchel, W., Volinsky, C., Johnson, T., Cortes, C. and Pregibon, D. (1999). Squashing flat files flatter. *Proceedings of the 5th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 6–15.
- Scott, S. L., Blocker, A. W., Bonassi, F. V., Chipman, H. A., George, E. I. and McCulloch, R. E. (2016). Bayes and big data: the consensus Monte Carlo algorithm. *International Journal of Management Science and Engineering Management*, 11(2), 78–88. doi:10.1080/17509653.2016.1142191
- Rue, H., Martino, S. and Chopin, N. (2009). Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society B*, 71(2), 319–392.

8 Appendix C – The complexity envelope

The table grades model features by whether exact aggregation applies. "Refused" means the model is rejected before any data is read, with the stated reason code; the per-row `flexybayes()` path fits it unchanged.

Table 4: Exact aggregation across increasing model complexity.

Model feature	Aggregates?	Why	Disposition
Factor fixed effects + random intercept	Yes – exact	Linear predictor is constant within a cell	Aggregated path
Gaussian / binomial / poisson, canonical link	Yes – exact	Finite additive sufficient statistic per cell	Aggregated path
Continuous fixed covariate	No	One cell per distinct value; K approaches N	Refused: continuous_cell_key_data_dependent
Factor-by-continuous interaction (indexed slope)	No	Numeric part varies within the cell	Refused: continuous_cell_key_data_dependent
Random slopes (x | g) / (x || g)	No	Contribution varies within the cell	Refused: random_slope_not_aggregatable
Structured covariance (us, ar1, spatial, GBLUP, pedigree)	No	Latent block is not cell-constant	Refused: structured_random_not_aggregatable
Smooth terms s(x), te(x, z)	No	Basis is observation-row-specific	Refused: smooth_term_not_aggregatable
Non-canonical link; gamma / beta / other family	No	No additive sufficient statistic (or not wired)	Refused: non_aggregatable_family / non_canonical_link
High-cardinality factor with little replication	Compresses little	K approaches N; aggregation gives no gain	Per-row path (not productive to aggregate)

Two boundaries are worth stating plainly. The aggregation is exact only for the likelihood; the posterior is still produced by an approximate inference engine (INLA's Laplace approximation or greta's Markov chain Monte Carlo), which is why `triangulate()` exists to cross-check the engines. And the method compresses only when $K \ll N$; the moment a design's cell count grows with the row count, exact aggregation offers nothing and the per-row path – or a labelled approximation – is the right tool.